

Stock Price Prediction using Price Features, Sentiment and Weighted Embeddings

This project predicts next-day stock direction (UP/DOWN) and returns using price data, sentiment (FinBERT), and embeddings. The goal is to evaluate whether combining structured financial features with unstructured text signals improves predictive performance.

Feature Engineering

News embeddings were incorporated in two ways: using only the latest news and using a weighted aggregation. The weighted approach is the main contribution of this project.

Each news item embedding is weighted based on sentiment confidence. Neutral news receives weight 1.0, while positive or negative news is amplified as $1 + \text{sentiment score}$. This allows the model to prioritize impactful news rather than treating all information equally.

Model

XGBoost was used as the main model. It builds decision trees sequentially, optimizing errors step by step. It works well for tabular data and can handle nonlinear interactions between price features, sentiment, and embeddings.

The dataset was split chronologically into train, validation, and test sets to avoid data leakage.

Classification Results

Metric	Value
Accuracy	0.5516
Precision	0.6267
Recall	0.6225
F1 Score	0.6246
ROC-AUC	0.5082
F1 Macro	0.5340
Balanced Accuracy	0.5340

The model performs slightly above random. It predicts upward movements better than downward ones, but overall discrimination power is weak.

Regression Results

MAE: 0.0146
RMSE: 0.0234
 R^2 : -0.10

Conclusion

Even with sentiment and embeddings, the model behaves close to random. This highlights the difficulty of predicting financial markets and suggests that additional signals or different modeling approaches are required.